

PDF 5 & 6: Web Application Architecture, HTTP & MVC

1. What is the default port for HTTP?

- a) 8080
- b) 443
- c) 80
- d) 21

Answer: c

2. HTTP is described as a stateless protocol because:

- a) It stores session information in cookies automatically
- b) The server does not automatically maintain conversational state between requests
- c) It does not support headers
- d) It only supports GET requests

Answer: b

3. Which HTTP method is used to request data from the server without changing it?

- a) POST
- b) DELETE
- c) PUT
- d) GET

Answer: d

4. What does an HTTP status code of 404 mean?

- a) Internal Server Error
- b) Unauthorized
- c) Not Found
- d) Moved Permanently

Answer: c

5. What does HTTP status code 200 indicate?

- a) Request redirected

- b) Request failed
- c) Server error
- d) Request was successful

Answer: d

6. Which HTTP method sends data in the request body (not in the URL)?

- a) GET
- b) POST
- c) HEAD
- d) OPTIONS

Answer: b

7. What is a Web Server's primary role?

- a) Execute database queries
- b) Serve static web pages using HTTP
- c) Manage user sessions
- d) Compile Java code

Answer: b

8. What does an Application Server do that a basic Web Server cannot?

- a) Serve HTML pages
- b) Support dynamic content generation, business logic execution, and resource management
- c) Accept HTTP requests
- d) Handle DNS lookups

Answer: b

9. What is CGI (Common Gateway Interface)?

- a) A database protocol
- b) A CSS specification
- c) A 1993 specification for HTTP servers to run programs and return dynamic content
- d) A Java framework

Answer: c

10. What is the main disadvantage of CGI over Servlets?

- a) CGI is more portable
- b) CGI spawns a new process per request (heavy), while Servlets use lightweight threads
- c) CGI uses Java while Servlets use Python
- d) CGI is more secure

Answer: b

11. In the MVC pattern, which component handles user input?

- a) Model
- b) View
- c) Controller
- d) Database

Answer: c

12. In MVC for web applications, the View is typically created by:

- a) Servlets
- b) Java Beans
- c) JSP pages
- d) JDBC

Answer: c

13. What does the Model in MVC do?

- a) Displays data to the user
- b) Handles HTTP requests
- c) Contains business logic and data representation
- d) Manages CSS styles

Answer: c

14. What are the 3 rules for a Java Bean?

- a) Public constructor with args, public variables, static methods
- b) Public no-argument constructor, no public instance variables, properties accessed via get/set methods
- c) Private constructor, public variables, no methods
- d) Abstract class, implements Serializable, has toString()

Answer: b

15. What is the Singleton design pattern?

- a) A pattern where a class can have unlimited instances
- b) A pattern ensuring only one instance of a class exists, with a global access point
- c) A pattern for creating multiple similar objects
- d) A pattern for database connections only

Answer: b

16. How is the single instance accessed in a Singleton pattern?

- a) By calling new ClassName()
- b) Via a public getInstance() method
- c) Via a public constructor
- d) By importing the class directly

Answer: b

17. In Model 2 (MVC) architecture, what acts as the Controller?

- a) JSP
- b) HTML
- c) Java Bean
- d) Servlet

Answer: d

18. What is the benefit of Model 2 (MVC) over Model 1 architecture?

- a) Model 1 allows better separation of concerns
- b) Model 2 separates flow control (Servlet) from presentation (JSP), making it more maintainable

- c) Model 2 uses fewer files
- d) Model 1 is more secure

Answer: b

19. Which Tomcat test URL confirms the server is running correctly?

- a) http://localhost/test
- b) http://localhost:8080
- c) http://server:80/tomcat
- d) http://127.0.0.1/admin

Answer: b

20. What is the HTTP HEAD method used for?

- a) To send data to the server
- b) To delete a resource
- c) To retrieve only the HTTP headers for a resource (no body)
- d) To update a resource

Answer: c

21. HTTP status codes in the range 500–599 indicate:

- a) Successful requests
- b) Redirections
- c) Client errors
- d) Server errors

Answer: d

22. What is HTTP keep-alive?

- a) A method to store sessions
- b) An HTTP 1.1 feature allowing multiple requests over the same connection
- c) A way to prevent server timeout
- d) A cookie that never expires

Answer: b

23. Which of the following is an Application Server?

- a) Apache HTTP Server
- b) NGINX
- c) IBM WebSphere (WAS)
- d) Lighttpd

Answer: c

24. What does the For a boolean "bar" property Java Bean respond to?

- a) getBar() and setBar()
- b) isBar() and setBar(boolean bar)
- c) fetchBar() and putBar()
- d) readBar() and writeBar()

Answer: b

25. Which HTTP status code means "Moved Permanently"?

- a) 302
- b) 404
- c) 200
- d) 301

Answer: d

26. What does a Web Server do when it receives a request for dynamic content?

- a) Returns an error
- b) Passes the request to the appropriate program/application server for processing
- c) Caches the response
- d) Rejects the request

Answer: b

27. A Java Bean property count of type int requires which getter?

- a) boolean isCount()

- b) String getCount()
- c) int getCount()
- d) void fetchCount()

Answer: c

28. Why is the Singleton pattern preferred over using static methods?

- a) Static methods use less memory
- b) Singletons can be subclassed and passed as arguments, offering more flexibility
- c) Static methods are thread-safe
- d) Singletons are faster

Answer: b

29. GET requests limit data to approximately how many characters in the URL?

- a) 500
- b) 2048
- c) 1024
- d) 256

Answer: c

30. The HTTP response status line contains:

- a) Only the status code
- b) The HTTP version, status code, and short description
- c) The response body only
- d) The request method and URL

Answer: b